HESSIAN DISCRIMINANTS AND AUTOMORPHISMS OF BLOW-UPS

OPEN AI

Abstract. Let $\pi\colon Y = \mathrm{Bl}_S X \to X$ be the blow-up of a smooth complex projective n-fold along a nonempty connected smooth centre of codimension $c \geq 2$. Write $W = \pi^* N^1(X)_{\mathbf{R}} \subset N^1(Y)_{\mathbf{R}}$ and let Δ_Y be the determinant of the Hessian bilinear form of the top intersection polynomial. We prove that W occurs in the Hessian discriminant with exact multiplicity $c-2$. For $c \geq 3$, it is therefore a linear component with finite orbit under $\mathrm{Aut}(Y)$. Its stabilizer is exactly the subgroup obtained by lifting $\mathrm{Aut}(X, S)$, and

$$[\mathrm{Aut}(Y) : \mathrm{Aut}(X, S)] \leq \left\lfloor \frac{(n-2)\rho(Y)}{c-2} \right\rfloor.$$

For a finite sequence of blow-ups along smooth centres of codimension at least three, a uniform power of every automorphism descends through the whole sequence. Consequently, if the initial smooth complex projective variety has finite automorphism group, then so does the final blow-up. Thus the examples asked for in Lesieutre–Litt's Question 1.8 do not occur when the initial variety is smooth over $\mathbf{C}$.

1. Introduction

Let X be a smooth complex projective variety and let Y be obtained from X by a sequence of ordinary blow-ups along smooth centres. A basic question is whether the birational modification can create new automorphisms. Bayraktar–Cantat obtained strong restrictions when the centres have sufficiently small dimension, and related questions were studied by Truong; see [1, 8]. Lesieutre–Litt proved a uniform descent theorem under the hypotheses

$$2r + 3 \leq n,$$

or $r + 3 \leq n$ together with polyhedrality of the exceptional nef cone.

for a centre of dimension r in an n-fold [6, Theorem 1.6]. They left open the intermediate range and asked whether blow-ups along smooth centres of codimension at least three can turn a finite automorphism group into an infinite one [6, Question 1.8].

The argument below uses only the top intersection polynomial on N^1. For a smooth projective n-fold Z, set

$$P_Z(h) = \int_Z h^n, \qquad q_h(u, v) = \int_Z h^{n-2} uv, \qquad \Delta_Z(h) = \det(q_h).$$

2020 *Mathematics Subject Classification.* 14J50, 14E05, 14C17.

Key words and phrases. blow-up, automorphism group, intersection form, Hessian discriminant, exceptional divisor.

Up to a nonzero scalar, Δ_Z is the determinant of the ordinary Hessian of P_Z. If $\pi\colon Y = \mathrm{Bl}_S X \to X$ has centre of codimension c, then the pullback space

$$W = \pi^* N^1(X)_{\mathbf{R}}$$

is a hyperplane in $N^1(Y)_{\mathbf{R}}$. The key point is the exact factorization order

$$\mathrm{ord}_W(\Delta_Y) = c - 2. \tag{1}$$

Thus codimension three is exactly the first codimension for which the blow-down is recorded by a linear component of the Hessian discriminant. Automorphisms preserve Δ_Y up to a nonzero scalar, hence permute these linear components with their multiplicities. The orbit of W is finite, and its stabilizer turns out to be exactly the subgroup lifted from automorphisms of the pair (X, S).

Intersection forms have long been used to constrain birational geometry. In dimension three, Bisi–Cascini–Tasin used the cubic intersection form to obstruct smooth blow-up presentations [2]. More recently, they studied the hypermatrix of the $(n-1)$st derivatives of the full intersection form; in particular, they derived a blow-up expansion and used low-rank exceptional classes to constrain divisorial contractions [3, Sections 4–5]. The invariant used here is different: it is the determinant of the ordinary Hessian on N^1, and the relevant locus is the pullback hyperplane W, rather than the exceptional class itself.

The exact multiplicity in (1) yields both an explicit index bound and a uniform iterate theorem. The latter removes the dimension and nef-cone hypotheses from Lesieutre–Litt's descent theorem when every centre has codimension at least three.

2. The Hessian discriminant

Let Z be a smooth complex projective n-fold, $n \geq 3$, and put

$$V_Z = N^1(Z)_{\mathbf{R}}, \qquad \rho(Z) = \dim_{\mathbf{R}} V_Z.$$

Choose a basis $D_1, \dots, D_\rho$ of V_Z. Define

$$q_h(u, v) = \int_Z h^{n-2} uv, \qquad M_Z(h) = \big(q_h(D_i, D_j)\big)_{i,j}, \qquad \Delta_Z(h) = \det M_Z(h). \tag{2}$$

If $h = \sum x_i D_i$, then

$$\det \mathrm{Hess}(P_Z) = \big(n(n-1)\big)^\rho \Delta_Z.$$

A change of basis multiplies Δ_Z by a nonzero square. Hence the zero hypersurface of Δ_Z, and the multiplicity of each real linear factor, are intrinsic.

Lemma 2.1. *The polynomial Δ_Z is nonzero and homogeneous of degree $(n-2)\rho(Z)$. Its zero hypersurface is preserved by $\mathrm{Aut}(Z)$, including the multiplicity of every irreducible factor.*

Proof. For an ample class a, the Hodge index theorem gives signature $(1, \rho(Z) - 1)$ for

$$(u, v) \longmapsto \int_Z a^{n-2} uv$$

on $N^1(Z)_{\mathbf{R}}$; see, for example, [4, Theorem 7.3.4]. Hence $\Delta_Z(a) \neq 0$. Each entry of $M_Z(h)$ is homogeneous of degree $n-2$, so $\deg \Delta_Z = (n-2)\rho(Z)$.

If $g \in \mathrm{Aut}(Z)$ and T is the matrix of g^* in the chosen basis, invariance of intersection numbers gives

$$T^{\mathrm{t}} M_Z(g^*h) T = M_Z(h).$$

Thus

$$\Delta_Z(g^*h) = (\det T)^{-2} \Delta_Z(h).$$

Pullback therefore preserves the zero hypersurface of Δ_Z and all factor multiplicities. □

3. The multiplicity of the blow-down hyperplane

Let

$$\pi \colon Y = \mathrm{Bl}_S X \longrightarrow X$$

be the blow-up of a smooth complex projective n-fold along a nonempty connected smooth centre S of codimension $c \geq 2$. Let E be the exceptional divisor. Then

$$N^1(Y)_{\mathbf{R}} = \pi^* N^1(X)_{\mathbf{R}} \oplus \mathbf{R}[E]. \tag{3}$$

Write

$$W = \pi^* N^1(X)_{\mathbf{R}}.$$

We first record the only intersection calculation needed below.

Lemma 3.1. *For $1 \leq j < c$ one has $\pi_* E^j = 0$, while*

$$\pi_* E^c = (-1)^{c-1}[S].$$

Proof. Let $p \colon E = \mathbf{P}(N_{S/X}) \to S$ and put $\xi = c_1(\mathcal{O}_E(1))$. Since $\mathcal{O}_E(E) = \mathcal{O}_E(-1)$,

$$E^j = i_*((-\xi)^{j-1}).$$

The projective bundle formula gives $p_*\xi^k = 0$ for $k < c-1$ and $p_*\xi^{c-1} = 1$. Pushing forward through $S \hookrightarrow X$ proves the claim; compare [5, Chapter 3]. □

Theorem 3.2 (Exact Hessian multiplicity)**.** *With the notation above,*

$$\boxed{\mathrm{ord}_W(\Delta_Y) = c - 2.}$$

Equivalently, if ℓ_W is a linear equation for W, then

$$\ell_W^{c-2} \mid \Delta_Y, \qquad \ell_W^{c-1} \nmid \Delta_Y.$$

In particular, W is a component of the Hessian discriminant if and only if $c \geq 3$.

Proof. Choose a basis $D_1, \dots, D_r$ of $N^1(X)_{\mathbf{R}}$ and use

$$\pi^* D_1, \dots, \pi^* D_r, E$$

as a basis of $N^1(Y)_{\mathbf{R}}$. Write

$$h = \pi^* a + tE.$$

Relative to this basis, the matrix of q_h has block form

$$M_Y(h) = \begin{pmatrix} A(t) & b(t) \\ b(t)^{\mathrm{t}} & d(t) \end{pmatrix}.$$

By lemma 3.1 and the projection formula,

$$A(t) = A(0) + O(t^c), \tag{4}$$

$$b(t) = O(t^{c-1}), \tag{5}$$

$$d(t) = (-1)^{c-1}\binom{n-2}{c-2}\left(\int_S (a|_S)^{n-c}\right) t^{c-2} + O(t^{c-1}). \tag{6}$$

The block estimates hold in the polynomial ring in the coordinates of a and t. Expanding the determinant by permutations therefore shows directly that

$$t^{c-2} \mid \det M_Y(\pi^* a + tE)$$

as a polynomial, not merely after specializing a.

Now take a ample. The Hodge index theorem gives $\det A(0) \neq 0$, and ampleness on S gives

$$\int_S (a|_S)^{n-c} > 0.$$

In the formal power series ring at $t = 0$,

$$\det M_Y(h) = \det A(t)\big(d(t) - b(t)^{\mathrm{t}} A(t)^{-1} b(t)\big).$$

The second term in the parentheses has order at least $2c - 2$, whereas (6) has a nonzero term of order $c - 2$. Therefore the coefficient of t^{c-2} in $\det M_Y(h)$ is nonzero for this ample a. The global divisibility by t^{c-2} is consequently exact. □

Remark 3.3 (The codimension-two boundary). For $c = 2$, theorem 3.2 gives multiplicity zero. More concretely, for ample a the block $A(0)$ is nondegenerate, the mixed block vanishes at $t = 0$, and

$$d(0) = -\int_S (a|_S)^{n-2} < 0.$$

Hence $\Delta_Y(\pi^* a) \neq 0$. This is exactly where the Hessian-hyperplane mechanism stops. Lesieutre–Litt exhibit codimension-two blow-ups for which uniform descent fails [6, Example 2.5].

4. Automorphisms of a single blow-up

Write

$$\mathrm{Aut}(X, S) = \{\varphi \in \mathrm{Aut}(X) : \varphi(S) = S\}.$$

Every element of $\mathrm{Aut}(X, S)$ lifts uniquely to Y.

Theorem 4.1 (Virtual recovery of the blow-down)**.** *Assume $c \geq 3$ and set*

$$M(Y, S) = \left\lfloor \frac{(n-2)\rho(Y)}{c-2} \right\rfloor.$$

Then the natural lift identifies $\mathrm{Aut}(X, S)$ *with the stabilizer of* W *in* $\mathrm{Aut}(Y)$. *In particular,*

$$[\mathrm{Aut}(Y) : \mathrm{Aut}(X, S)] \leq M(Y, S). \tag{7}$$

Moreover, if

$$L(Y, S) = \mathrm{lcm}(1, 2, \ldots, M(Y, S)),$$

then for every $g \in \mathrm{Aut}(Y)$ *the iterate* $g^{L(Y,S)}$ *descends to an automorphism of* X *preserving* S.

Proof. By theorem 3.2, the linear form defining W divides Δ_Y with multiplicity $c - 2$. By lemma 2.1, $\mathrm{Aut}(Y)$ permutes the linear factors having this multiplicity. There are at most

$$\left\lfloor \frac{\deg \Delta_Y}{c-2} \right\rfloor = M(Y, S).$$

Thus the stabilizer

$$G_W = \{g \in \mathrm{Aut}(Y) : g^* W = W\}$$

has index at most $M(Y, S)$.

It remains to identify G_W. Let $g \in G_W$ and choose an ample divisor A on X. If $C \subset Y$ is an integral curve contracted by π, then

$$A \cdot \pi_* g_* [C] = (g^* \pi^* A) \cdot C = 0,$$

because $g^* \pi^* A \in W$. If $\pi(g(C))$ were a curve, ampleness would make the left side positive. Hence g sends every contracted curve to a contracted curve.

Each nontrivial fibre of π is $\mathbf{P}^{c-1}$, whose closed points are joined by lines. Therefore πg is constant on every fibre. Let Γ be the reduced image of

$$(\pi, \pi g) \colon Y \longrightarrow X \times X.$$

This map is proper, so its image is closed. The first projection $q \colon \Gamma \to X$ is proper. For every $x \in X$, all points of $\pi^{-1}(x)$ have the same image in Γ; since $Y \to \Gamma$ is surjective, the underlying topological space of $q^{-1}(x)$ consists of one point. Thus q has finite fibres and is finite [7, Tag 02LS]. Over $X \setminus S$ it is the graph of the morphism $\pi g \pi^{-1}$, so q is birational. The variety Γ is integral, and X is normal; hence q is an isomorphism [7, Tag 0AB1]. The second projection therefore gives a morphism $\bar{g} \colon X \to X$ satisfying

$$\pi g = \bar{g} \pi.$$

Applying the same argument to g^{-1} gives a morphism $\overline{g^{-1}} \colon X \to X$. Since π is surjective,

$$\bar{g}\, \overline{g^{-1}} = \overline{g^{-1}}\, \bar{g} = \mathrm{id}_X,$$

so $\bar{g} \in \mathrm{Aut}(X)$.

For $y \in E$, choose a line in the fibre of $E \to S$ through y. Its image under g is contracted by π. Every curve contracted by π is contained in E, because π is an isomorphism over $X \setminus S$. Hence $g(E) \subseteq E$; since both are irreducible divisors, $g(E) = E$. Therefore

$$\bar{g}(S) = \pi g(E) = S.$$

If $\bar{g} = \mathrm{id}_X$, then g is the identity on $Y \setminus E$, and hence on Y. Therefore descent gives an injection $G_W \hookrightarrow \mathrm{Aut}(X, S)$. Conversely, every automorphism of (X, S) lifts uniquely to the blow-up by its universal property, and the lift preserves W. Hence

$$G_W \cong \mathrm{Aut}(X, S),$$

which proves (7).

Finally, the orbit of W under a fixed g has length at most $M(Y, S)$. Therefore g^m stabilizes W for some $1 \le m \le M(Y, S)$, and $g^{L(Y,S)}$ does so uniformly for every g. □

5. Sequences of blow-ups

A blow-up along a finite disjoint union of smooth connected centres is the same as the successive blow-up of the components. We therefore state the result for connected centres.

Theorem 5.1 (Uniform descent)**.** *Let*

$$Y = X_m \xrightarrow{\pi_m} X_{m-1} \longrightarrow \cdots \xrightarrow{\pi_1} X_0 = X$$

be a finite sequence of smooth complex projective n-folds, where π_i is the blow-up along a nonempty connected smooth centre $S_i \subset X_{i-1}$ of codimension $c_i \ge 3$. Put

$$M_i = \left\lfloor \frac{(n-2)\rho(X_i)}{c_i - 2} \right\rfloor, \qquad L_i = \mathrm{lcm}(1, \dots, M_i), \qquad N = \prod_{i=1}^{m} L_i.$$

Then:

(i) *there is a finite-index subgroup $G \le \mathrm{Aut}(Y)$ that descends faithfully to $\mathrm{Aut}(X)$;*

(ii)
$$[\mathrm{Aut}(Y) : G] \le \prod_{i=1}^{m} M_i;$$

(iii) *for every $g \in \mathrm{Aut}(Y)$, the iterate g^N descends to an automorphism of X.*

Consequently, if $\mathrm{Aut}(X)$ is finite, then $\mathrm{Aut}(Y)$ is finite.

Proof. For each i, theorem 4.1 gives a subgroup $G_i \le \mathrm{Aut}(X_i)$ of index at most M_i together with an injective descent homomorphism

$$\delta_i \colon G_i \longrightarrow \mathrm{Aut}(X_{i-1}).$$

Define recursively $H_0 = \mathrm{Aut}(X_0)$ and

$$H_i = \delta_i^{-1}(H_{i-1}) \subset G_i.$$

Then

$$[G_i : H_i] = [\delta_i(G_i) : \delta_i(G_i) \cap H_{i-1}] \le [\mathrm{Aut}(X_{i-1}) : H_{i-1}].$$

Induction gives

$$[\mathrm{Aut}(X_i) : H_i] \le \prod_{j=1}^{i} M_j.$$

Taking $G = H_m$ proves (i) and (ii), and faithfulness follows from the single-step faithfulness at every stage.

For (iii), start with $g_m = g$. By theorem 4.1, $g_m^{L_m}$ descends to some $g_{m-1} \in \mathrm{Aut}(X_{m-1})$. Then $g_{m-1}^{L_{m-1}}$ descends to X_{m-2}, and so on. After m steps, g^N descends to X. If $\mathrm{Aut}(X)$ is finite, then G is finite by injectivity; since G has finite index, so is $\mathrm{Aut}(Y)$. □

Corollary 5.2. *Let X be smooth and complex projective. For any finite sequence of blow-ups of X along smooth centres of codimension at least three, Lesieutre–Litt's uniform-iterate conclusion holds with no additional dimension inequality and no polyhedral-nef-cone hypothesis. In particular, no such sequence produces an infinite automorphism group from a finite one; this rules out the examples sought in* [6, Question 1.8] *when the initial variety is smooth over* $\mathbf{C}$.

Remark 5.3. The proof is numerical except for the nondegeneracy of the Hodge-index form used in lemma 2.1. No dynamical Mordell–Lang input is required. We have kept the statement over $\mathbf{C}$ in order to isolate the argument from additional characteristic-dependent hypotheses.

AI assistance disclosure

GPT-6 Astra and GPT-5.6 Sol were used during parts of the investigation, drafting, and checking of this note.

References

[1] T. Bayraktar and S. Cantat, *Constraints on automorphism groups of higher dimensional manifolds*, J. Math. Anal. Appl. **405** (2013), no. 1, 209–213.

[2] C. Bisi, P. Cascini, and L. Tasin, *A remark on the Ueno–Campana's threefold*, Michigan Math. J. **65** (2016), no. 3, 567–572.

[3] C. Bisi, P. Cascini, and L. Tasin, *Forms and complex manifolds*, EMS Surv. Math. Sci. (2025), published online first, DOI: 10.4171/EMSS/109.

[4] M. A. A. de Cataldo, *Lectures on the Hodge theory of projective manifolds*, arXiv:math/0504561 (2005).

[5] W. Fulton, *Intersection Theory*, 2nd ed., Springer, 1998.

[6] J. Lesieutre and D. Litt, *Dynamical Mordell–Lang and automorphisms of blow-ups*, Algebraic Geometry **6** (2019), no. 1, 1–25.

[7] The Stacks Project Authors, *The Stacks Project*, `https://stacks.math.columbia.edu`, Tags 02LS and 0AB1.

[8] T. T. Truong, *On automorphisms of blowups of projective manifolds*, arXiv:1301.4957 (2013).